

STA1381 – Pengantar Sains Data Pertemuan-3

Data Management

Anang Kurnia
Cici Suhaeni
Septian Rahardiantoro

2026

Outline

Manajemen Data

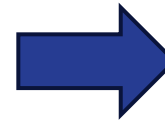
- Menciptakan peubah baru melalui Regular Expression
- Preprocessing data text

Manajemen Data

- Pada data web scraping yang sudah diperoleh, akan dilakukan proses manajemen data
- Manajemen data dimaksudkan untuk:
 - ✓ Menyesuaikan format peubah (numerik, teks)
 - ✓ Menyusun peubah-peubah tertentu

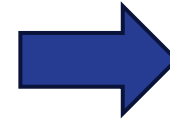
1. Perhatikan data hasil web scraping Data Mobil Bekas

```
[1] "93.142 km, Automatic, Tangerang Selatan, "  
[2] "33.827 km, Automatic, Jakarta Utara, "  
[3] "54.801 km, Automatic, Tangerang Selatan, "  
[4] "25.160 km, Manual, Tangerang Selatan, "  
[5] "95.389 km, Automatic, Jakarta Utara, "  
[6] "140.903 km, Automatic, Tangerang Selatan, "  
[7] "103.106 km, Automatic, Bekasi, "  
[8] "46.694 km, Automatic, Jakarta Utara, "  
[9] "24.619 km, Manual, Tangerang Selatan, "  
[10] "22.670 km, Automatic, Tangerang Selatan, "  
[11] "94.946 km, Automatic, Bekasi, "  
[12] "104.055 km, Automatic, Tangerang Selatan, "  
[13] "63.203 km, Automatic, Jakarta Utara, "  
[14] "79.013 km, Automatic, Tangerang Selatan, "  
[15] "81.128 km, Manual, Jakarta Utara, "  
[16] "44.889 km, Automatic, Jakarta Utara, "  
[17] "98.215 km, Automatic, Tangerang Selatan, "  
[18] "73.975 km, Automatic, Jakarta Utara, "
```



Dapat diekstrak
menjadi 3 peubah

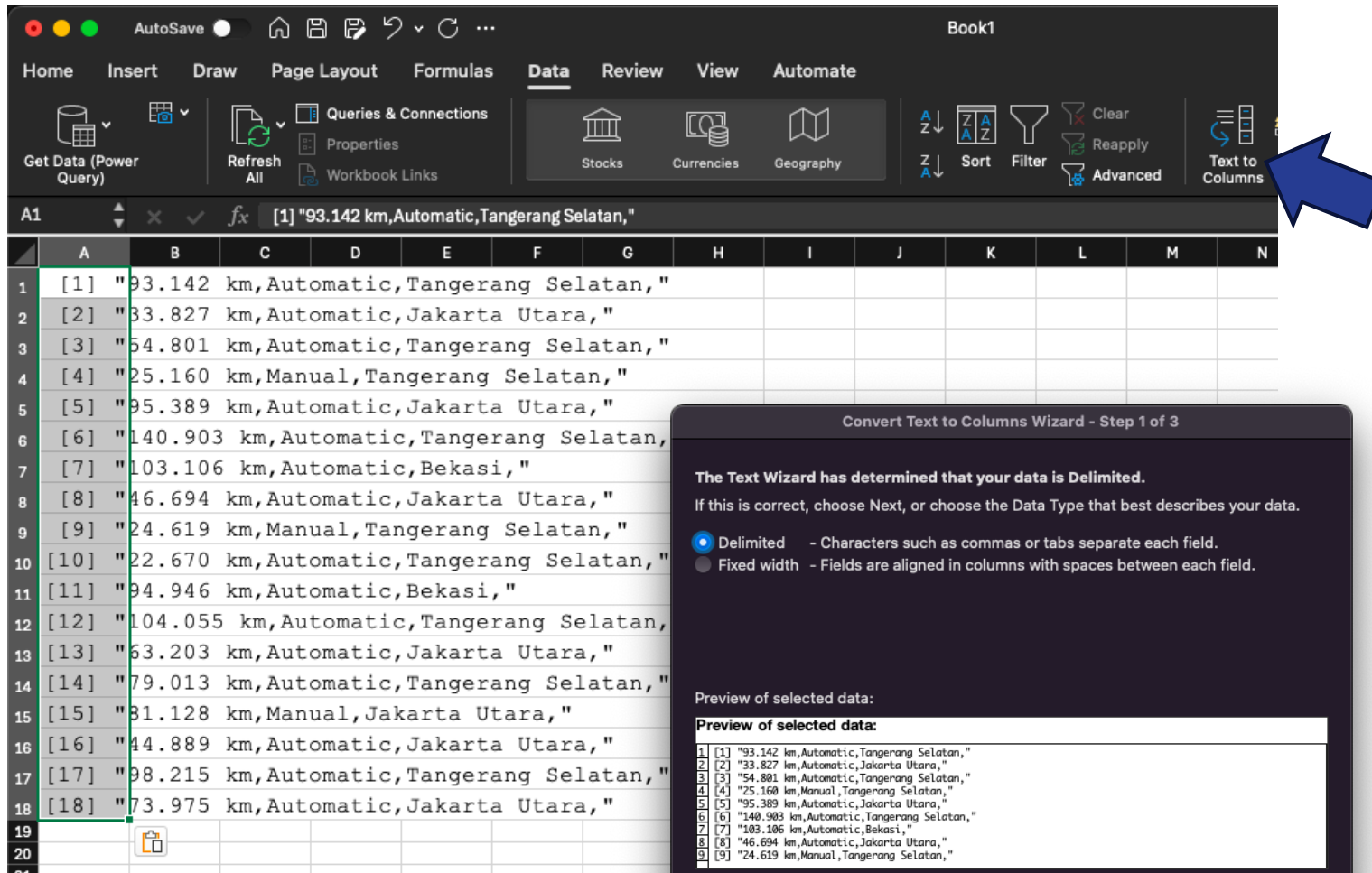
```
[1] "93.142 km, Automatic, Tangerang Selatan, "
[2] "33.827 km, Automatic, Jakarta Utara, "
[3] "54.801 km, Automatic, Tangerang Selatan, "
[4] "25.160 km, Manual, Tangerang Selatan, "
[5] "95.389 km, Automatic, Jakarta Utara, "
[6] "140.903 km, Automatic, Tangerang Selatan, "
[7] "103.106 km, Automatic, Bekasi, "
[8] "46.694 km, Automatic, Jakarta Utara, "
[9] "24.619 km, Manual, Tangerang Selatan, "
[10] "22.670 km, Automatic, Tangerang Selatan, "
[11] "94.946 km, Automatic, Bekasi, "
[12] "104.055 km, Automatic, Tangerang Selatan, "
[13] "63.203 km, Automatic, Jakarta Utara, "
[14] "79.013 km, Automatic, Tangerang Selatan, "
[15] "81.128 km, Manual, Jakarta Utara, "
[16] "44.889 km, Automatic, Jakarta Utara, "
[17] "98.215 km, Automatic, Tangerang Selatan, "
[18] "73.975 km, Automatic, Jakarta Utara, "
```



Jarak	Transmisi	Wilayah
93.142 km	Automatic	Tangerang Selatan
33.827 km	Automatic	Jakarta Utara
54.801 km	Automatic	Tangerang Selatan
25.160 km	Manual	Tangerang Selatan
95.389 km	Automatic	Jakarta Utara
140.903 km	Automatic	Tangerang Selatan
103.106 km	Automatic	Bekasi
46.694 km	Automatic	Jakarta Utara
24.619 km	Manual	Tangerang Selatan
22.670 km	Automatic	Tangerang Selatan
94.946 km	Automatic	Bekasi
104.055 km	Automatic	Tangerang Selatan
63.203 km	Automatic	Jakarta Utara
79.013 km	Automatic	Tangerang Selatan
81.128 km	Manual	Jakarta Utara
44.889 km	Automatic	Jakarta Utara
98.215 km	Automatic	Tangerang Selatan
73.975 km	Automatic	Jakarta Utara

Bagaimana caranya?

Ekstraksi peubah → dengan Ms. Excel



The screenshot shows the Microsoft Excel interface with the **Data** tab selected. The **Text to Columns** button in the Data Tools group is highlighted with a blue arrow. The worksheet contains a list of data entries in column A, such as "[1] 93.142 km, Automatic, Tangerang Selatan,".

Menu **Data** → Text to Columns

Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is **Delimited**.
If this is correct, choose Next, or choose the Data Type that best describes your data.

☒ **Delimited** - Characters such as commas or tabs separate each field.
☐ **Fixed width** - Fields are aligned in columns with spaces between each field.

Preview of selected data:

	Preview of selected data:
[1]	"93.142 km, Automatic, Tangerang Selatan,"
[2]	"33.827 km, Automatic, Jakarta Utara,"
[3]	"54.801 km, Automatic, Tangerang Selatan,"
[4]	"25.160 km, Manual, Tangerang Selatan,"
[5]	"95.389 km, Automatic, Jakarta Utara,"
[6]	"140.903 km, Automatic, Tangerang Selatan,"
[7]	"103.106 km, Automatic, Bekasi,"
[8]	"46.694 km, Automatic, Jakarta Utara,"
[9]	"24.619 km, Manual, Tangerang Selatan,"

Cancel < Back **Next >** Finish

Pilih **Delimited**, Next >

Convert Text to Columns Wizard - Step 2 of 3

This screen lets you set the delimiters your data contains.

Delimiters

☒ **Tab** ☐ Treat consecutive delimiters as one
☐ Semicolon
☒ **Comma** Text qualifier: "
☐ Space
☐ Other:

Preview of selected data:

	Preview of selected data:
[1]	"93.142 km Automatic Tangerang Selatan"
[2]	"33.827 km Automatic Jakarta Utara"
[3]	"54.801 km Automatic Tangerang Selatan"
[4]	"25.160 km Manual Tangerang Selatan"
[5]	"95.389 km Automatic Jakarta Utara"
[6]	"140.903 km Automatic Tangerang Selatan"
[7]	"103.106 km Automatic Bekasi"
[8]	"46.694 km Automatic Jakarta Utara"
[9]	"24.619 km Manual Tangerang Selatan"

Cancel < Back **Next >** Finish

Pilih **Delimiters** yang sesuai
(dalam hal ini "Comma"), Next >

Jarak	Transmisi	Wilayah
93.142 km	Automatic	Tangerang Selatan
33.827 km	Automatic	Jakarta Utara
54.801 km	Automatic	Tangerang Selatan
25.160 km	Manual	Tangerang Selatan
95.389 km	Automatic	Jakarta Utara
140.903 km	Automatic	Tangerang Selatan
103.106 km	Automatic	Bekasi
46.694 km	Automatic	Jakarta Utara
24.619 km	Manual	Tangerang Selatan
22.670 km	Automatic	Tangerang Selatan
94.946 km	Automatic	Bekasi
104.055 km	Automatic	Tangerang Selatan
63.203 km	Automatic	Jakarta Utara
79.013 km	Automatic	Tangerang Selatan
81.128 km	Manual	Jakarta Utara
44.889 km	Automatic	Jakarta Utara
98.215 km	Automatic	Tangerang Selatan
73.975 km	Automatic	Jakarta Utara

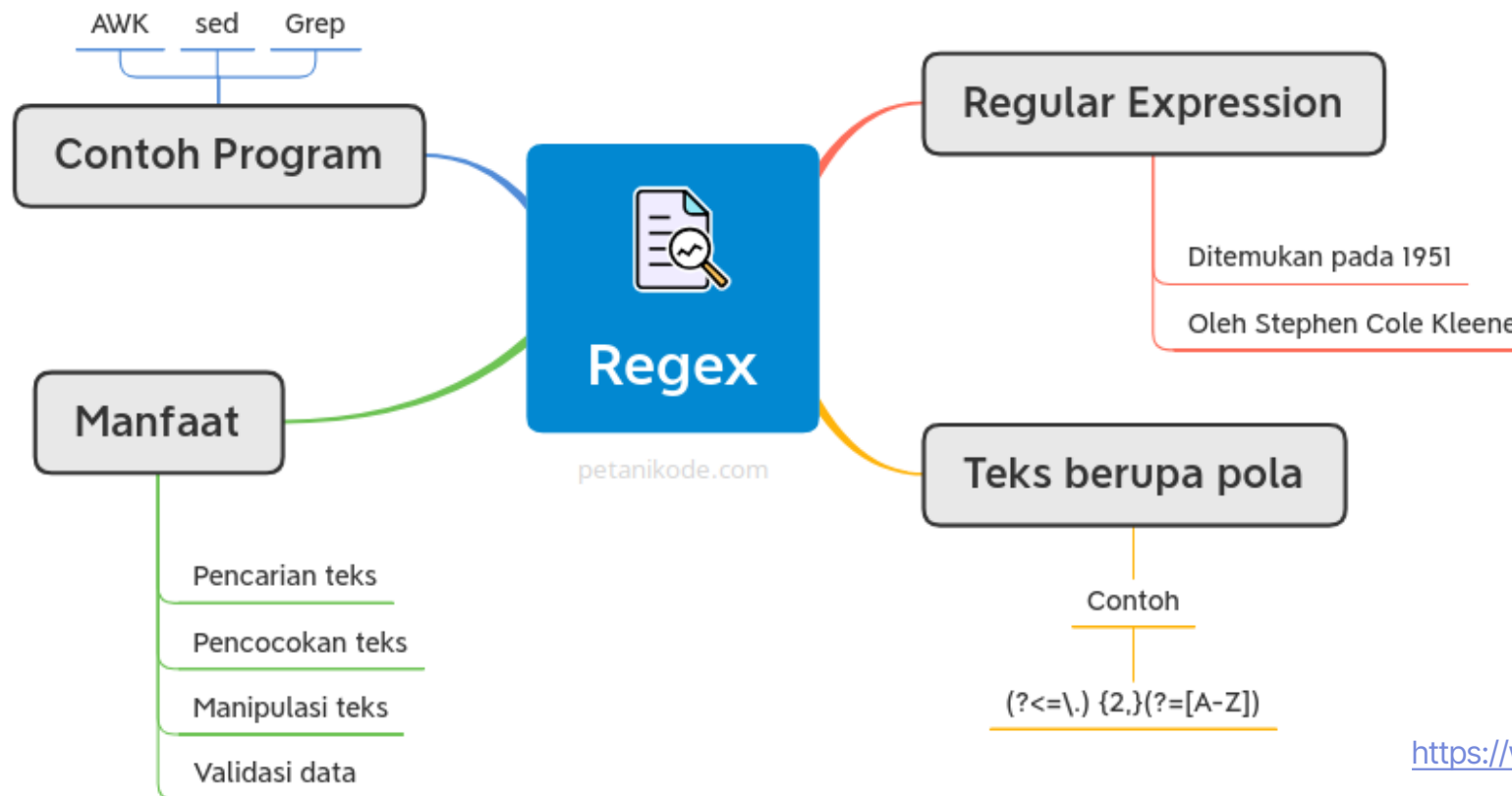


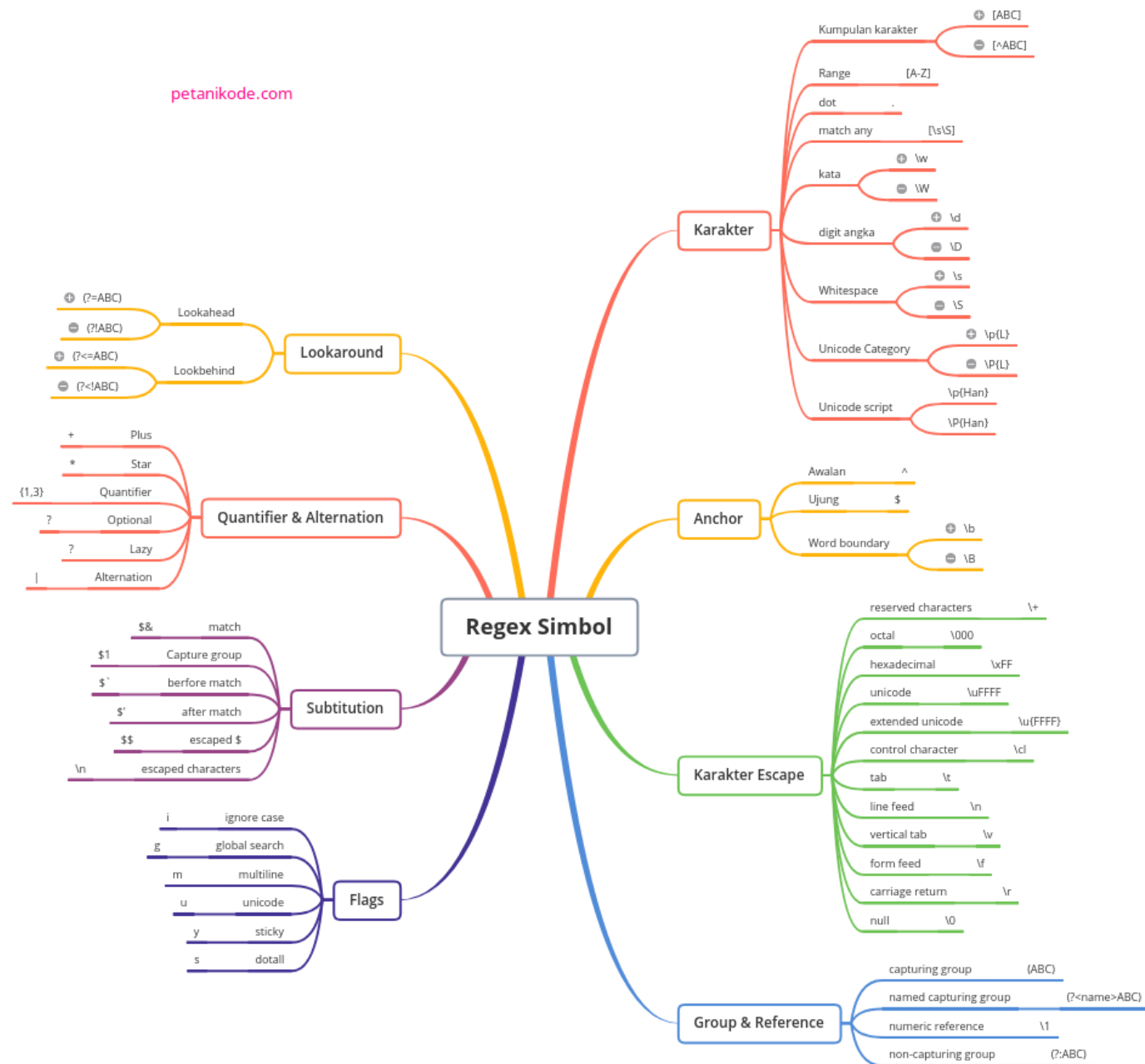
Numerik???

Gunakan Regular Expression (Regex)

Regular Expression (RegEx)

→ Susunan string yang mendefinisikan pola pencarian sehingga dapat membantu kita untuk melakukan matching (pencocokan), locate (pencarian), dan manipulasi teks.







Beberapa simbol yang sering digunakan:

()	→ untuk grouping;
	→ simbol untuk logika or (atau alternatif);
[a-z]	→ untuk mencari karakter dari huruf a sampai z;
[A-Z]	→ untuk mencari karakter dari huruf A sampai Z;
[0-9]	→ Untuk mencari angka 0 sampai 9;
\w	→ untuk mencari kata;
\d	→ untuk mencari angka;



Dapat diterapkan di R



Regular Expression (RegEx) di R

- R mendukung regular expression
- Dapat pula menggunakan package tambahan, seperti **stringr** (recommended), **stringi**, dll
- Beberapa fungsi di antaranya:

Fungsi	stringr	R Base
Mengembalikan semua element string yang cocok	<code>str_subset()</code>	<code>grep()</code>
Memeriksa keberadaan pola dalam string	<code>str_detect()</code>	<code>grepl()</code>
Ekstraksi pola yang cocok dari suatu string	<code>str_extract()</code> , <code>str_extract_all()</code>	<code>regexpr()</code> , <code>regmatches()</code> , <code>grep()</code>
Menemukan posisi dari pola dalam suatu string	<code>str_locate()</code>	<code>regexpr()</code>
Menemukan semua posisi dari pola dalam suatu string	<code>str_locate_all()</code>	<code>gregexpr()</code>
Mengganti pola yang cocok dari suatu string	<code>str_replace()</code>	<code>sub()</code>
Mengganti semua yang cocok dg pola dari suatu string	<code>str_replace_all()</code>	<code>gsub()</code>

Ilustrasi di R

```
##ilustrasi regex
library(stringr)
perusahaan <- c("Roti Bangsawan", "08128712", "RB@email.id",
               "roti_bang29@gmail.com", "F3823KSB")
str_subset(perusahaan, 'a')
str_detect(perusahaan, '[0-9]')
str_extract_all(perusahaan, '[0-9]')
str_extract_all(perusahaan, '[0-9]+')
str_extract(perusahaan, '[a-z]+')
str_extract_all(perusahaan, '[a-z]+')
str_extract_all(perusahaan, '[A-Za-z]+')
str_replace(perusahaan, 'com|id', 'net')
str_replace_all(perusahaan, '[^a-z0-9]', '#')
str_remove_all(perusahaan, '[._@]+')
```



```
> str_subset(perusahaan, 'a')
[1] "Roti Bangsawan"      "RB@email.id"
[3] "roti_bang29@gmail.com"
> str_detect(perusahaan, '[0-9]')
[1] FALSE  TRUE  FALSE  TRUE  TRUE
> str_extract_all(perusahaan, '[0-9]')
[[1]]
character(0)

[[2]]
[1] "0" "8" "1" "2" "8" "7" "1" "2"

[[3]]
character(0)

[[4]]
[1] "2" "9"

[[5]]
[1] "3" "8" "2" "3"
```

```
> str_extract_all(perusahaan, '[0-9]+')
[[1]]
character(0)

[[2]]
[1] "08128712"

[[3]]
character(0)

[[4]]
[1] "29"

[[5]]
[1] "3823"

> str_extract(perusahaan, '[a-z]+')
[1] "oti"    NA      "email" "roti"  NA
```



```
> str_extract_all(perusahaan, '[a-z]+')  
[[1]]
```

```
[1] "oti"      "angsawan"
```

```
[[2]]  
character(0)
```

```
[[3]]  
[1] "email" "id"
```

```
[[4]]  
[1] "roti"  "bang"  "gmail" "com"
```

```
[[5]]  
character(0)
```

```
> str_extract_all(perusahaan, '[A-Za-z]+')  
[[1]]
```

```
[1] "Roti"      "Bangsawan"
```

```
[[2]]  
character(0)
```

```
[[3]]  
[1] "RB"      "email" "id"
```

```
[[4]]  
[1] "roti"  "bang"  "gmail" "com"
```

```
[[5]]  
[1] "F"      "KSB"
```

```
> str_replace(perusahaan, 'com|id', 'net')
```

```
[1] "Roti Bangsawan"      "08128712"
```

```
[3] "RB@email.net"        "roti_bang29@gmail.net"
```

```
[5] "F3823KSB"
```

```
> str_replace_all(perusahaan, '[^a-z0-9]', '#')
```

```
[1] "#oti##angsawan"      "08128712"
```

```
[3] "###email#id"         "roti#bang29#gmail#com"
```

```
[5] "#3823###"
```

```
> str_remove_all(perusahaan, '[._@]')
```

```
[1] "Roti Bangsawan"      "08128712"
```

```
"RBemailid"
```

```
[4] "rotibang29gmailcom"  "F3823KSB"
```




```
[1] "93.142 km, Automatic, Tangerang Selatan,"  
[2] "33.827 km, Automatic, Jakarta Utara,"  
[3] "54.801 km, Automatic, Tangerang Selatan,"  
[4] "25.160 km, Manual, Tangerang Selatan,"  
[5] "95.389 km, Automatic, Jakarta Utara,"  
[6] "140.903 km, Automatic, Tangerang Selatan,"  
[7] "103.106 km, Automatic, Bekasi,"  
[8] "46.694 km, Automatic, Jakarta Utara,"  
[9] "24.619 km, Manual, Tangerang Selatan,"  
[10] "22.670 km, Automatic, Tangerang Selatan,"  
[11] "94.946 km, Automatic, Bekasi,"  
[12] "104.055 km, Automatic, Tangerang Selatan,"  
[13] "63.203 km, Automatic, Jakarta Utara,"  
[14] "79.013 km, Automatic, Tangerang Selatan,"  
[15] "81.128 km, Manual, Jakarta Utara,"  
[16] "44.889 km, Automatic, Jakarta Utara,"  
[17] "98.215 km, Automatic, Tangerang Selatan,"  
[18] "73.975 km, Automatic, Jakarta Utara,"
```



Regular Expression (RegEx)

```
dt1 <- read.csv("data1.csv")  
str(dt1)  
dt1  
  
library(stringr)  
x <- dt1$x  
  
split_words <- strsplit(x, ",\\s*")  
  
max_length <- max(sapply(split_words, length))  
split_words_df <- data.frame(t(sapply(split_words, function(x)  
{  
  length(x) <- max_length  
  return(x)  
})), stringsAsFactors = FALSE)  
  
colnames(split_words_df) <- c("Jarak_Tempuh", "Tipe", "Asal")  
print(split_words_df)
```

`",\\s*` berarti bahwa string akan dipisahkan di setiap koma, dan karakter spasi yang mengikuti koma akan diabaikan (dihapus).



```
> print(split_words_df)
```

	Jarak_Tempuh	Tipe	Asal
1	93.142 km	Automatic	Tangerang Selatan
2	33.827 km	Automatic	Jakarta Utara
3	54.801 km	Automatic	Tangerang Selatan
4	25.160 km	Manual	Tangerang Selatan
5	95.389 km	Automatic	Jakarta Utara
6	140.903 km	Automatic	Tangerang Selatan
7	71.135 km	Automatic	Jakarta Utara
8	46.694 km	Automatic	Jakarta Utara
9	24.619 km	Manual	Tangerang Selatan
10	22.670 km	Automatic	Tangerang Selatan
11	94.946 km	Automatic	Bekasi
12	104.055 km	Automatic	Tangerang Selatan
13	63.203 km	Automatic	Jakarta Utara
14	79.013 km	Automatic	Tangerang Selatan
15	81.128 km	Manual	Jakarta Utara
16	44.889 km	Automatic	Jakarta Utara
17	98.215 km	Automatic	Tangerang Selatan
18	73.975 km	Automatic	Jakarta Utara



Numerik???

```
library(stringr)
#remove dot
w1 <-
str_remove_all(split_words_df[,1], "[.]")
#ekstrak angka
w2 <- str_extract_all(w1, "[0-9]+")
#dibuat format numerik
w3 <- as.numeric(w2) #sudah ok
#data ok
df_oke <- split_words_df
df_oke$Jarak_Tempuh <- w3
df_oke$Tipe <- as.factor(df_oke$Tipe)
df_oke$Asal <- as.factor(df_oke$Asal)
df_oke
str(df_oke)
```

2. Perhatikan data hasil web scraping Data Handphone Android

web-scraper-order	web-scraper-start-url	nama_hp	harga_hp		
1692063197-1	https://www.blibli.com/cari/handphone%20android	Xiaomi Redmi 9C Smartphone [4GB/ 64GB]	Rp1.589.000		
1692063197-2	https://www.blibli.com/cari/handphone%20android	Asus Zenfone 3 ZE520KL Smartphone [32GB/ 4GB]	Rp1.299.000		
1692063197-3	https://www.blibli.com/cari/handphone%20android	Xiaomi Poco M3 Pro 5G Smartphone [4GB/64GB]	Rp2.250.000		
1692063197-4	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy S21 FE 8/256GB RAM 8 ROM 256 Handphone Hp Smartphone Android Garansi Resmi	Rp7.099.000		
1692063197-5	https://www.blibli.com/cari/handphone%20android	Vivo Y15S 3/64 GB RAM 3 ROM 64GB Handphone Hp Smartphone Android Original Murah Garansi Resmi	Rp1.480.000		
1692063197-6	https://www.blibli.com/cari/handphone%20android	OPPO A31 Handphone Android - Black [4/128]	Rp2.547.902		
1692063197-7	https://www.blibli.com/cari/handphone%20android	Advan NasaPlus 2GB 16GB Smartphone Android10 Garansi Resmi	Rp845.000		
1692063197-8	https://www.blibli.com/cari/handphone%20android	OPPO A57 4/64 GB Smartphone Android Original Garansi Resmi	Mulai		Rp2.399.000
1692063197-9	https://www.blibli.com/cari/handphone%20android	Advan NasaPlus 2GB 16GB Smartphone Android10 Garansi Resmi Deep Blue	Rp845.000		
1692063197-10	https://www.blibli.com/cari/handphone%20android	Hp xiomni not 9	Rp2.100.000		
1692063197-11	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A04e 3/32 GB RAM 3GB 32GB Smartphone Android	Rp1.137.000		
1692063197-12	https://www.blibli.com/cari/handphone%20android	Infinix Hot 30i 8/128 RAM 8GB ROM 128GB 8 128 GB HP Android	Rp1.475.000		
1692063197-13	https://www.blibli.com/cari/handphone%20android	POCO C40 3/32 GB - Resmi	Rp1.258.000		
1692063197-14	https://www.blibli.com/cari/handphone%20android	Handphone OPPO A37	Rp700.000		
1692063197-15	https://www.blibli.com/cari/handphone%20android	XIAOMI REDMI 10A RAM 3GB - 64GB GARANSI RESMI	Mulai		Rp1.149.000
1692063197-16	https://www.blibli.com/cari/handphone%20android	Xiaomi MI 11T Pro 12/256 GB RAM 12 ROM 256GB Handphone Smartphone Android Original Garansi Resmi	Rp7.899.000		
1692063197-17	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A13 4/128 6/128 4GB 6GB 128GB RAM 4 6 ROM 128 GB HP Smartphone Android	Mulai		Rp2.500.000
1692063197-18	https://www.blibli.com/cari/handphone%20android	handphone samsung A12 6/128 resmi sein	Rp2.750.000		
1692063197-19	https://www.blibli.com/cari/handphone%20android	Advan Nasa Pro 2 GB 32 GB Smartphone Android 11	Rp799.000		
1692063197-20	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A73 5G 8/256 8GB 256GB RAM 8 ROM 256 GB Smartphone Android	Rp8.518.000		
1692063197-21	https://www.blibli.com/cari/handphone%20android	ALDO A9 SMARTPHONE RAM 3GB INTERNAL 16GB ANDROID	Rp680.000		
1692063197-22	https://www.blibli.com/cari/handphone%20android	HP Oppo A53S murah terbaru 2022 OPPO A53 S 5G RAM 8 + 256 GB Original New Baru Smartphone Android 6.52in HP OPPO Murah garansi 1 tahun	Rp2.459.000		
1692063197-23	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy S21 FE 8/256 GB RAM 8 ROM 256 Handphone Hp Smartphone Android Garansi Resmi	Rp7.149.000		
1692063197-24	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A04e Smartphone [3/32GB]	Rp1.500.000		
1692063197-25	https://www.blibli.com/cari/handphone%20android	Samsung A14 4/128	Rp2.399.000		
1692063197-26	https://www.blibli.com/cari/handphone%20android	Oppo Reno 8T 5G 8/256GB	Rp6.199.000		
1692063197-27	https://www.blibli.com/cari/handphone%20android	OPPO A96 8GB/256GB Garansi Resmi 1 Tahun	Rp3.231.000		
1692063197-28	https://www.blibli.com/cari/handphone%20android	Vivo V25E 8/128 Garansi resmi vivo	Rp3.699.000		
1692063197-29	https://www.blibli.com/cari/handphone%20android	Huawei Y6 II LTE Smartphone - Black [16GB/ 2GB]	Rp3.849.000		
1692063197-30	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A54 8/128GB	Rp5.599.000		
1692063197-31	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A54 8/256GB	Rp5.999.000		
1692063197-32	https://www.blibli.com/cari/handphone%20android	SAMSUNG A03 CORE 2/32 GB [GARANSI RESMI]	Rp1.199.000		
1692063197-33	https://www.blibli.com/cari/handphone%20android	Xiaomi Redmi 9C Smartphone [4GB/ 64GB]	Rp1.625.000		
1692063197-34	https://www.blibli.com/cari/handphone%20android	Vivo Y16 - 3GB/32GB	Rp1.799.000		
1692063197-35	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A04s Smartphone [4/64GB]	Mulai		Rp1.799.000
1692063197-36	https://www.blibli.com/cari/handphone%20android	Tecno Pova 4 (8GB + 128GB) Garansi Resmi	Rp2.269.000		
1692063197-37	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A04s Smartphone [4/64GB]	Rp2.487.500		
1692063197-38	https://www.blibli.com/cari/handphone%20android	Samsung Galaxy A04s Smartphone [4/64GB]	Mulai		Rp3.799.000
1692063197-39	https://www.blibli.com/cari/handphone%20android	Xiaomi Redmi 9C Smartphone [4GB/ 64GB]	Rp1.110.000		
1692063197-40	https://www.blibli.com/cari/handphone%20android	Vivo Y16 Smartphone [4GB/128GB]	Rp1.110.000		
1692063197-41	https://www.blibli.com/cari/handphone%20android	REALME C31 RAM 3/32 GB GARANSI RESMI	Rp1.110.000		
1692063197-42	https://www.blibli.com/cari/handphone%20android	Redmi 9A 2/32GB	Rp1.110.000		
1692063197-43	https://www.blibli.com/cari/handphone%20android	Infinix Note 7 Lite Smartphone [4GB- 64GB]	Rp2.100.000		
1692063197-44	https://www.blibli.com/cari/handphone%20android	Xiaomi redmi 9T RAM 4/64 GB garansi resmi	Rp2.475.000		
1692063197-45	https://www.blibli.com/cari/handphone%20android	Samsung M23 5G 6/128 garansi resmi	Rp3.309.800		

Dapat di ekstrak menjadi beberapa peubah

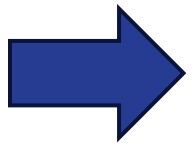
Belum numerik

Penyesuaian peubah harga untuk menjadi peubah numerik

➡ Numerik??? **Gunakan Regular Expression (Regex)**

Kembali ke data

Penyesuaian peubah harga untuk menjadi peubah numerik



Numerik???

```
#data
data.hp <- read.csv("/Users/gad/Documents/SRO/Pengantar
Sains Data/Ganjil 24 25/contohhp.csv",
                  header=T, sep=";")

str(data.hp)

harga.hp <- data.hp$hargahp
length(harga.hp)
head(harga.hp)

library(stringr)
#remove dot
h1 <- str_remove_all(harga.hp, "[.]")
#ekstrak angka
h2 <- str_extract_all(h1, "[0-9]+")
#dibuat format numerik
h3 <- as.numeric(h2) #sudah ok
head(h3)
```



Ekstraksi peubah

nama_hp	harga_hp		
Xiaomi Redmi 9C Smartphone [4GB/ 64GB]	Rp1.589.000		
Asus Zenfone 3 ZE520KL Smartphone [32GB/ 4GB]	Rp1.299.000		
Xiaomi Poco M3 Pro 5G Smartphone [4GB/ 64GB]	Rp2.250.000		
Samsung Galaxy S21 FE 8/256GB RAM 8 ROM 256 Handphone Hp Smartphone Android Garansi Resmi	Rp7.099.000		
Vivo Y15S 3/64 GB RAM 3 ROM 64GB Handphone Hp Smartphone Android Original Murah Garansi Resmi	Rp1.480.000		
OPPO A31 Handphone Android - Black [4/128]	Rp2.547.902		
Advan NasaPlus 2GB 16GB Smartphone Android10 Garansi Resmi	Rp845.000		
OPPO A57 4/64 GB Smartphone Android Original Garansi Resmi	Mulai		Rp2.399.000
Advan NasaPlus 2GB 16GB Smartphone Android10 Garansi Resmi Deep Blue	Rp845.000		
Hp xiami not 9	Rp2.100.000		
Samsung Galaxy A04e 3/32 GB RAM 3GB 32GB Smartphone Android	Rp1.137.000		
Infinix Hot 30i 8/128 RAM 8GB ROM 128GB 8 128 GB HP Android	Rp1.475.000		
POCO C40 3/32 GB - Resmi	Rp1.258.000		
Handphone OPPO A37	Rp700.000		
XIAOMI REDMI 10A RAM 3GB - 64GB GARANSI RESMI	Mulai		Rp1.149.000
Xiaomi MI 11T Pro 12/256 GB RAM 12 ROM 256GB Handphone Smartphone Android Original Garansi Resmi	Rp7.899.000		
Samsung Galaxy A13 4/128 6/128 4GB 6GB 128GB RAM 4 6 ROM 128 GB HP Smartphone Android	Mulai		Rp2.500.000
handphone samsung A12 6/128 resmi sein	Rp2.750.000		
Advan Nasa Pro 2 GB 32 GB Smartphone Android 11	Rp799.000		
Samsung Galaxy A73 5G 8/256 GB 256GB RAM 8 ROM 256 GB Smartphone Android	Rp8.518.000		
ALDO A9 SMARTPHONE RAM 3GB INTERNAL 16GB ANDROID	Rp680.000		
HP Oppo A53S murah terbaru 2022 OPPO A53 S 5G RAM 8 + 256 GB Original New Baru Smartphone Android 6.52in HP OPPO Murah garansi 1 tahun	Rp2.459.000		
Samsung Galaxy S21 FE 8/256 GB RAM 8 ROM 256 Handphone Hp Smartphone Android Garansi Resmi	Rp7.149.000		
Samsung Galaxy A04e Smartphone [3/32GB]	Rp1.500.000		
Samsung A14 4/128	Rp2.399.000		
Oppo Reno 8T 5G 8/256GB	Rp6.199.000		
OPPO A96 8GB/256GB Garansi Resmi 1 Tahun	Rp3.231.000		
Vivo V25E 8/128 Garansi resmi vivo	Rp3.699.000		
Huawei Y6 II LTE Smartphone - Black [16GB/ 2GB]	Rp3.849.000		
Samsung Galaxy A54 8/128GB	Rp5.599.000		
Samsung Galaxy A54 8/256GB	Rp5.999.000		
SAMSUNG A03 CORE 2/32 GB [GARANSI RESMI]	Rp1.199.000		
Xiaomi Redmi 9C Smartphone [4GB/ 64GB]	Rp1.625.000		
Vivo Y16 - 3GB/32GB	Rp1.799.000		
Samsung Galaxy A04s Smartphone [4/64GB]	Mulai		Rp1.799.000
Tecno Pova 4 (8GB + 128GB) Garansi Resmi	Rp2.269.000		
Samsung Galaxy A13 6/128GB	Rp2.487.500		
Samsung Galaxy A23 5G 6/128GB	Mulai		Rp3.799.000
Xiaomi Redmi 9A Smartphone [32 GB/ 2 GB]	Rp1.110.000		
Vivo Y16 Smartphone [4GB/128GB]	Rp1.899.000		
REALME C31 RAM 3/32 GB GARANSI RESMI	Rp1.470.000		
Redmi 9A 2/32GB	Rp1.500.500		
Infinix Note 7 Lite Smartphone [4GB- 64GB]	Rp2.100.000		
Xiaomi redmi 9T RAM 4/64 GB garansi resmi	Rp2.475.000		
Samsung M23 5G 6/128 garansi resmi	Rp3.309.800		

Memori Internal

RAM

Nama merek HP

Bagaimana Caranya ???

Ekstraksi peubah **Nama merek HP**

```
##ekstraksi peubah
#ekstraksi nama merek HP
nama.hp <- data.hp$namahp
head(nama.hp)

#identifikasi list nama HP
list.hp1 <- str_extract(nama.hp, '[A-Za-z]+')
hp2 <- as.factor(list.hp1)
summary(hp2)

list.hp2 <- str_extract(nama.hp, '\\b(?:SMARTPHONE\\b)\\w+\\b')
hp3 <- as.factor(list.hp2)
summary(hp3)

list.hp3 <- str_extract(nama.hp, '\\b(?:SMARTPHONE\\b|[Hh][Pp]\\b)\\w+\\b')
hp4 <- as.factor(list.hp3)
summary(hp4)

##dan seterusnya
```

Bagaimana Ekstraksi peubah
RAM dan Memori Internal?



Preprocessing Data Text



Scrapping data dulu...

- Misalkan kita akan scrapping data komentar youtube
- Hal yang harus dilakukan :
 - Mendapatkan Youtube API
 - Scrapping menggunakan Python



Bagaimana mendapatkan Youtube API?

2 Cara Mendapatkan YouTube API Key

2.1 Langkah 1: Akses Google Developers Console

2.2 Langkah 2: Buat Proyek Baru

2.3 Langkah 3: Aktifkan YouTube Data API

2.4 Langkah 4: Buat Kredensial API Key

2.5 Langkah 5: YouTube API Key Sudah

Didapatkan

<https://idwebhost.com/blog/youtube-api-key-adalah/>

<https://developers.google.com/youtube/v3/getting-started?hl=id>



Kode python untuk scrapping

Install libraries

```
!pip install google-api-python-client openpyxl pandas
```

Import libraries

```
from googleapiclient.discovery import build  
import pandas as pd
```

Initialize YouTube API

```
API_KEY = "TULISKAN YOUTUBE API KAMU"  
youtube = build("youtube", "v3", developerKey=API_KEY)
```

Function to fetch videos

```
def search_videos(keyword, max_results=2):  
    request = youtube.search().list(  
        part="snippet", q=keyword, type="video", maxResults=max_results)  
    response = request.execute()  
    videos = [{  
        "title": item["snippet"]["title"],  
        "description": item["snippet"]["description"],  
        "publish_date": item["snippet"]["publishedAt"],  
        "video_id": item["id"]["videoId"],  
        "channel": item["snippet"]["channelTitle"],  
        "channel_id": item["snippet"]["channelId"]  
    } for item in response.get("items", [])]  
    return videos
```



Function to fetch comments

```
def fetch_comments(video_id):
    comments = []
    try:
        request = youtube.commentThreads().list(part="snippet", videoId=video_id, maxResults=100)
        response = request.execute()
        while response:
            for item in response.get("items", []):
                comment = item["snippet"]["topLevelComment"]["snippet"]
                comments.append({
                    "author": comment["authorDisplayName"],
                    "comment": comment["textDisplay"],
                    "published_at": comment["publishedAt"],
                    "like_count": comment["likeCount"]
                })
            if "nextPageToken" in response:
                request = youtube.commentThreads().list(
                    part="snippet", videoId=video_id, pageToken=response["nextPageToken"], maxResults=100)
                response = request.execute()
            else:
                break
    except Exception as e:
        print(f"Error fetching comments for video {video_id}: {e}")
    return comments
```



```
# Fetch video data
```

```
keyword = "demo DPR"
```

```
videos = search_videos(keyword, max_results=2)
```

```
# Fetch comments and channel details for videos
```

```
all_comments = []
```

```
video_info = []
```

```
for video in videos:
```

```
    # Fetch comments for each video
```

```
    comments = fetch_comments(video["video_id"])
```

```
    for comment in comments:
```

```
        all_comments.append({
            "author": comment["author"],
            "comment": comment["comment"],
            "published_at": comment["published_at"],
            "like_count": comment["like_count"],
            "channel": video["channel"],
            "title": video["title"]
        })
```



Prepare video details

```
video_info.append({  
    "title": video["title"],  
    "description": video["description"],  
    "publish_date": video["publish_date"],  
    "url": f"https://www.youtube.com/watch?v={video['video_id']}",  
    "channel": video["channel"],  
    "channel_id": video["channel_id"]  
})
```

Create DataFrames

```
comments_df = pd.DataFrame(all_comments)  
videos_df = pd.DataFrame(video_info)
```

Save DataFrames to Excel

```
comments_df.to_excel('/kaggle/working/youtube_comments.xlsx', index=False)  
videos_df.to_excel('/kaggle/working/youtube_videos.xlsx', index=False)
```

```
print("Data comments saved to /kaggle/working/youtube_comments.xlsx")  
print("Data videos saved to /kaggle/working/youtube_videos.xlsx")
```


Ada 2 output:

- comment
- video

Create DataFrames

```
comments_df = pd.DataFrame(all_comments)
videos_df = pd.DataFrame(video_info)
```

Save DataFrames to Excel

```
comments_df.to_excel('/kaggle/working/youtube_comments.xlsx', index=False)
videos_df.to_excel('/kaggle/working/youtube_videos.xlsx', index=False)
```

```
print("Data comments saved to /kaggle/working/youtube_comments.xlsx")
print("Data videos saved to /kaggle/working/youtube_videos.xlsx")
```

	author	comment	published_at	like_count	channel	title
1	@suliamegarani683	Yg bisa membubarkan DPR memang rakyat, tp nyatanya	2025-09-03T14:47:0	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
2	@mr.donqel7770	TURUNKAN PRABOWO ✖ LANTIK ANIS BASWEDAN	2025-09-03T14:29:3	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
3	@alena2270	Jangan kasih kendor menyuarkan hapus tunjangan DP	2025-09-03T14:17:5	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
4	@didinislam2	Harusnya jika Presiden Outsourcing Ketua DPR juga Out	2025-09-03T14:10:2	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
5	@AlvinCristian-f7b	Bismillah ya Allah semoga tidak ada demo terjadi di tem	2025-09-03T13:53:5	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
6	@Saidin2024	Hapus DPR	2025-09-03T13:49:1	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
7	@diegokenzie7712	🔴🔴🔴🔴🔴🔴🔴🔴 hancurlah Indonesia	2025-09-03T13:36:5	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
8	@murasad6823	Saudara saudaraku silakan berdemo sampaikan aspiras	2025-09-03T13:35:0	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
9	@tasyaredmi3437	Ini di mana??	2025-09-03T13:34:5	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
10	@sategule7860	Alhamdulillah aman	2025-09-03T13:30:2	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
11	@hendrahermawan85	Ga ush ada DPR LAH	2025-09-03T13:26:3	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
12	@Tarmono-cg6pv	Semoga pejuang2 rakyat diberi keselamatan dan kesehat	2025-09-03T13:21:3	1	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
13	@teguhwiyono3719	HORMAT ACUNG JEMPOL 10 DEMO DAMAI INTELEK ME	2025-09-03T13:06:5	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
14	@mangkuncir258	bbrkn dpr dprd lurah n camat hkm mti koruptor n r	2025-09-03T13:06:2	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
15	@WasroPemalang	Bubarkan dpr,	2025-09-03T13:00:3	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
16	@Baguszord01	Mantap	2025-09-03T12:48:4	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025
17	@KireinaFelicia	Begini dong Kalo demo yang tertib ga anarkis	2025-09-03T12:37:3	0	BeritaSatu	🔴 BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025

Output 1 : comment



Output scrappingnya

title	description	publish_date	url	channel	channel_id
● BREAKING NEWS Situasi Terkini Demo DPR, 3 September 2025	Situasi Terkini Demo DPR, 3 September 2025 #BreakingNews #Demo #DPR #SituasiTerkini #Beritasatu ...	2025-09- 03T04:52:45Z	https://www.youtube.com/watch?v=9zpSVMzPm9A	BeritaSatu	UCqLsfkQSM0yfyGvONAGWd3Q
BREAKING NEWS - Situasi Terkini Demo di Depan DPR Jakarta 3 September 2025	BREAKING NEWS - Situasi Terkini Demo di Depan DPR Jakarta 3 September 2025 #dpr #demo #breakingnews Sahabat ...	2025-09- 03T04:04:25Z	https://www.youtube.com/watch?v=qhVEyMAiF60	KOMPASTV	UC5BMIWZe9isJXLZZWPWvBlg

Output 2 : video

<https://www.youtube.com/watch?v=9zpSVMzPm9A>

<https://www.youtube.com/watch?v=qhVEyMAiF60>



Bagaimana preprocessing data text?

Bagaimana preprocessing data text berbahasa Indonesia?

PREPROCESSING STEPS

01

Text Cleaning

- Removing URLs
- Removing Punctuation and Special Characters
- Removing Numbers
- Expanding Contractions
- Case Conversion (Lowercase/Uppercase)
- Handling Emojis
- Correcting Words (Repeating character/spelling)
- Removing Extra Whitespaces

02

Tokenization

03

Stop words Removal

Library yang dibutuhkan:

- re atau regex
- string
- NLTK
- Sastrawi
- indoNLP

04

Stemming/ Lemmatization



2. TOKENIZATION

Tokenization adalah proses membagi teks panjang, seperti paragraf atau kalimat, menjadi unit-unit yang lebih kecil dan lebih mudah dikelola serta diproses

Biasanya, kita melakukan **tokenization** sebelum atau sesudah **menghapus karakter dan simbol yang tidak diperlukan** dari data.

Pilihan ini tergantung pada **masalah yang ingin diselesaikan** dan **jenis data yang digunakan**.

✓ Library yang digunakan adalah NLTK (Natural Language Toolkit).

💡 Selain NLTK, library lain yang sering digunakan untuk Tokenization adalah `spaCy` dan

`TextBlob` . 🚀



2. TOKENIZATION

Tokenization dapat dilakukan dalam dua cara utama:

- 1 **Sentence Tokenization** → Memecah teks menjadi kalimat
- 2 **Word Tokenization** → Memecah teks menjadi kata

```
import nltk
from nltk.tokenize import sent_tokenize

# Download model yang diperlukan (hanya perlu sekali)
nltk.download('punkt')

# Contoh teks
text = "Tokenization is an important step in NLP. It helps break text into smaller units."

# Memecah teks menjadi kalimat
sentences = sent_tokenize(text)

# Output hasil
print("Hasil Sentence Tokenization:")
print(sentences)
```

Hasil Sentence Tokenization:

```
['Tokenization is an important step in NLP.', 'It helps break text into smaller units.']
```



2. TOKENIZATION

```
from nltk.tokenize import word_tokenize

# Contoh teks
text = "Tokenization helps NLP models understand text."

# Memecah teks menjadi kata
words = word_tokenize(text)

# Output hasil
print("Hasil Word Tokenization:")
print(words)
```

Hasil Word Tokenization:

```
['Tokenization', 'helps', 'NLP', 'models', 'understand', 'text', '.']
```




3. STOPWORDS REMOVAL

Stopwords adalah kata-kata yang sering muncul dalam suatu bahasa tetapi **tidak memiliki makna yang signifikan** dalam analisis teks.

Contohnya adalah kata-kata seperti **"the", "is", "and", "in", "at"**, yang biasanya tidak menambah nilai penting dalam pemrosesan bahasa alami (NLP).

a, an, and, are, as, at, be, but, by, for, if, in, into, is, it, no, not, of, on, or, such, that, the, their, then, there, these, they, this, to, was, will, with

✓ Mengapa Stopwords Perlu Dihapus?

- Stopwords sering kali tidak memiliki makna penting dalam analisis teks.
- Menghapus stopwords membantu model NLP fokus pada informasi yang lebih relevan.

3. STOPWORDS REMOVAL

📌 Sebelum Stopwords Removal:

"This is an example of a sentence that contains many stopwords in the English language, and it is not very meaningful for text analysis."

```
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

# Download stopwords jika belum ada
nltk.download('stopwords')
nltk.download('punkt')
# Contoh teks dengan banyak stopwords
text = "This is an example of a sentence that contains many stopwords in the English language, and it is not very meaningful"
# Tokenization
words = word_tokenize(text)
# Menghapus stopwords
stop_words = set(stopwords.words('english'))
filtered_text = [word for word in words if word.lower() not in stop_words]
# Output hasil
print("Setelah Stopwords Removal:", " ".join(filtered_text))
```

Setelah Stopwords Removal: example sentence contains many stopwords English language , meaningful text analysis .



4. STEMMING / LEMMATIZATION

Stemming is a simple and practical approach that involves cutting off the ends of words with the intention of obtaining the correct root form.

Eg:
automate, automatic, automation → automat

Lemmatization aims to remove inflectional endings only and to return the base or dictionary form of a word, which is known as the '*lemma*'.

Eg:
car, cars, cars', car's → car
am, is, are → be



Setelah data text selesai di preproses, selanjutnya:

- Text representation
- Visualization
- Modeling

Terima Kasih



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